

Mandatory Guidelines for Reinforcement Learning Mini-Project Assessment

Application-independent template for presentation, implementation, demonstration and viva

1. Overall Assessment Structure

The Mini-Project Assessment is divided into TWO PARTS. Part I evaluates CO1, CO2 and CO3; Part II evaluates CO4 and CO5. Together, both parts constitute the 20-mark mini-project assessment.

Part	COs Evaluated	Focus	Expected Evidence	Marks
Part I	CO1, CO2, CO3	RL understanding, problem formulation, algorithm selection, implementation and demonstration.	Presentation, RL formulation, source code, execution, model/policy where applicable, demonstration and viva.	10
Part II	CO4, CO5	Performance/convergence analysis, complexity/regret/sample-complexity considerations, and exploration–exploitation.	Experimental analysis, performance metrics, learning/convergence evidence, exploration strategy and viva.	10
Total	CO1–CO5	Complete mini-project evaluation.	Evidence from both parts.	20

2. Part I — CO1, CO2 and CO3 (10 Marks)

CO	Expected Learning Evidence	Assessment Focus	Marks
CO1	Define the key features of RL that distinguish it from AI and non-interactive ML.	Problem understanding and RL justification; explain why the selected problem is suitable for RL and distinguish RL from supervised/non-interactive ML.	3
CO2	Decide whether the problem should be formulated as RL; if yes, define state, action, dynamics and reward and select an appropriate algorithm.	RL problem formulation and algorithm selection.	2
CO3	Implement common RL algorithms in code.	Code implementation and demonstration of a working RL algorithm.	5

3. Part I — Mandatory Presentation Template

Slide	Mandatory Content
1. Title	Project title, team members/IDs, course, faculty and academic year.
2. Problem Statement	Define the problem, objective, environment and decision that must be made.
3. Why RL?	Explain interaction, sequential decisions, actions, feedback/reward and why RL is appropriate.
4. RL vs Other ML	Briefly compare RL with supervised/non-interactive ML and/or fixed rule-based approaches.
5. Agent–Environment Model	Identify agent, environment, observations/states, actions, rewards and interaction loop.
6. System Architecture	Show the flow: input/observation → state → RL agent → action → environment → reward.
7. State Space	Define every state variable, representation and range/discretization. Give an example state.
8. Action Space	List all actions and explain what each action means.
9. Reward Function	Give the mathematical/rule-based reward and justify positive and negative rewards.
10. Algorithm Selection	Name the RL algorithm and justify it using state/action characteristics and project requirements.
11. Core Equation	Show the main update/objective equation actually used by the implementation.
12. Implementation	Show software modules, data flow and important hyperparameters.
13. Demo	Demonstrate one complete state → action → reward → next-state cycle.
14. Results	Show actual learning curves and relevant task-performance metrics.
15. Analysis	Interpret learning, reward trend, success rate and stability.
16. Limitations/Future Work	State current limitations and realistic improvements.
17. Conclusion	Summarize formulation, implementation and demonstration; map evidence to CO1–CO3.

4. CO1 — Problem Understanding & RL Justification (3 Marks)

Mandatory: The project must be genuinely formulated as an RL problem. Using AI, deep learning, computer vision or a neural network alone does not make a project an RL project.

- Clearly define the problem and decision-making objective.
- Identify the agent and environment.
- Explain what the agent observes.
- Define the possible actions.
- Explain how the environment responds to actions.
- Define the feedback/reward mechanism.
- Explain why the problem involves sequential decision-making.
- Briefly compare the formulation with supervised/non-interactive ML.

Minimum comparison

Aspect	Supervised / Conventional ML	Reinforcement Learning
Learning signal	Label or target value	Reward / return
Interaction	Normally no action–environment loop	Agent interacts with environment
Decision	Prediction	Action selection
Temporal structure	Often independent samples	Sequential decisions
Objective	Prediction loss	Expected cumulative return

5. CO2 — RL Problem Formulation (2 Marks)

State: Define the information available to the agent at time t ; specify variables, representation and range where relevant.

Action: Define the complete set of decisions available to the agent and state whether actions are discrete or continuous.

Reward: Define desirable and undesirable outcomes and justify reward values.

Dynamics: Explain how an action changes the environment and produces the next state.

Objective: State the goal in terms of cumulative/discounted return or task performance.

Algorithm: Select an appropriate algorithm and justify it based on state/action characteristics, exploration needs and computational constraints.

Where appropriate, express the problem as an MDP: $M = (S, A, P, R, \gamma)$, where S is the state space, A the action space, P the transition dynamics, R the reward function and γ the discount factor.

6. CO3 — Implementation & Demonstration (5 Marks)

Required Evidence	What the student must show
Working implementation	At least one RL algorithm is implemented and executes successfully.
State handling	Show where the state is constructed and passed to the agent.
Action selection	Show how the agent selects an action using the policy/value function.
Environment interaction	Show how the action produces the next state and reward.
Learning/update	Show the code/module responsible for updating the value function or policy.
Training evidence	Show episodes/iterations, logs and important hyperparameters.
Demonstration	Show at least one complete end-to-end execution.
Model/policy	Show the trained model, policy or learned parameters where applicable.

Recommended live-demo sequence

Step	Demonstration
1	Start the environment/application.
2	Show the current observation/state.
3	Show available actions and selected action.
4	Execute the action and show the environment response.
5	Show reward and next state.
6	Show the learning/update step.

Step	Demonstration
7	Repeat or show the trained policy.
8	Display learning curve and final performance metrics.

7. Part II — CO4 and CO5 (10 Marks)

Part II extends the project beyond basic implementation. Students must analyse the behaviour and performance of the RL algorithm and demonstrate understanding of exploration and exploitation.

CO	Focus	Mandatory Evidence
CO4	Evaluate RL performance using regret, sample complexity, computational complexity, performance and convergence concepts.	Learning curves; performance metrics; convergence/stability analysis; computational/resource discussion; sample-efficiency or regret analysis where applicable.
CO5	Understand and apply exploration–exploitation strategies.	Exploration mechanism; exploitation mechanism; parameter/strategy used; and analysis of its effect on learning.

8. Part II — Mandatory Analysis

- **Performance:** Report suitable metrics such as success rate, cumulative reward, return, completion time or domain-specific measures.
- **Learning behaviour:** Plot reward/return against episode or iteration and explain the trend.
- **Convergence:** Discuss whether performance stabilizes, improves, oscillates or fails to converge.
- **Complexity:** Discuss computational cost, training time, memory, model size or relevant resources.
- **Sample efficiency:** Explain how much interaction/data the agent requires to learn effectively.
- **Regret:** Where meaningful, define or estimate cumulative regret and interpret it.
- **Exploration–exploitation:** Explain how the agent balances trying new actions with selecting known good actions.
- **Parameter sensitivity:** Discuss the effect of ϵ , learning rate, γ , entropy coefficient or other relevant parameters.

9. Suggested Results Table

Metric	Baseline / Initial	Final / Proposed	Interpretation
Average reward	_____	_____	Did cumulative performance improve?
Success rate	_____	_____	Did task success improve?
Convergence episode	_____	_____	When did performance stabilize?
Training time	_____	_____	Computational requirement
Exploration parameter	_____	_____	How exploration affected learning
Regret / sample efficiency	_____	_____	Where applicable

10. Team Project — Individual Contribution

For team projects, the application may be common, but each member must have a clearly identifiable technical responsibility. If multiple RL agents/modules are used, each member should be able to explain the assigned agent, including its state, action, reward, algorithm, implementation and results.

Member	Assigned Agent / Module	Individual Technical Evidence
Member 1	_____	State + action + reward + algorithm + code + results
Member 2	_____	State + action + reward + algorithm + code + results
Member 3	_____	State + action + reward + algorithm + code + results
Member 4	_____	State + action + reward + algorithm + code + results

11. Viva — Minimum Preparation

1. Why is your problem suitable for RL?
2. What is the difference between your observation and state?
3. What is the agent trying to optimize?
4. Why did you choose this reward function?
5. What happens after an action is selected?
6. How does the agent learn from reward?
7. Why is the selected algorithm appropriate?
8. How does your implementation handle exploration and exploitation?
9. What does the discount factor do?

10. How do you know whether the agent has learned?
11. What evidence supports convergence or improvement?
12. What are the computational/sample-efficiency limitations?
13. Which part of the project did you personally implement?

12. Final Mandatory Submission Checklist

- Presentation covering Part I (CO1–CO3) and Part II (CO4–CO5)
- Clear problem statement and RL justification
- Agent, environment, state, action, reward and dynamics defined
- Algorithm selection justified
- Working source code and execution evidence
- End-to-end demonstration
- Trained model/policy where applicable
- Learning curves and performance metrics
- Convergence/performance/complexity analysis for CO4
- Exploration–exploitation strategy and analysis for CO5
- Individual contribution clearly documented
- Limitations and future work

Golden Rule: The evaluator should be able to trace the complete chain: Problem → RL Justification → State → Action → Reward → Algorithm → Code → Learning → Performance → Exploration/Exploitation → Conclusion.